Challenges of Spatio-Temporal Trajectory Data Use: Focus Group Findings from the 1st International Summer School on Data Science for Mobility

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ABSTRACT

The fast development of wireless location acquisition technologies has led to a significant increase in the availability of mobility data, specifically spatio-temporal trajectory data, which includes information about the movements (locations) of objects over time. This data has proven valuable for a wide range of applications, including predicting travel patterns, discovering routes, analyzing social interactions, and managing resources in urban environments. However, using trajectory datasets can also be challenging, particularly in different countries. This article aims to explore the challenges of using trajectory datasets, specifically through the use of focus group discussions. Focus groups are a qualitative method of gaining a deeper understanding of a particular topic, and have not been previously used to examine the challenges related to trajectory datasets. The information gathered through these discussions is augmented by a review of existing literature for a comprehensive understanding of data challenges.

CCS CONCEPTS

• Spatial-temporal systems • Global positioning systems • Real-time systems

KEYWORDS

Spatio-temporal datasets, Trajectory analysis, Challenges, Focus group, Data privacy, Data ethics

1 Introduction

The rapid advancement in collecting spatio-temporal trajectory data has been driven by many factors, including the widespread use of GPS-enabled smartphone devices, increased investment in sustainable transportation and smart city technology, and the growth of the Internet of Things (IoT) and sensor-based connected devices [1]. This has led to the development of more sophisticated tools and methods for analyzing spatio-temporal data for different research applications, such as real-time traffic management, urban planning, environmental monitoring, marketing, and advertising, human behavior and activity analysis, disaster management and response, telecommunication network optimization, and Health and Safety (H&S) of people [2]. However, despite the numerous benefits of trajectory datasets, challenges exist regarding their use. These challenges include open questions and concerns related to dataset construction, dataset preprocessing, dataset privacy, dataset-sharing strategies to avoid data leaks, and the use of synthetic and private datasets. To better understand these challenges, this paper aims to explore the perspectives and experiences concerning the use of trajectory datasets through a focus group study that involves a diverse group of individuals from academia and industry who come from different countries.

A focus group study involves a focus on particular issues, with a planned group of people, participating in an interactive discussion [3]. These discussions fall under the category of the qualitative research tradition, which is an interactive conversation between predetermined participants, led by a moderator, and focusing on a particular set of issues [3]. The objective of a focus group discussion is to obtain a broad range of views on the research topic over a specific time period and to create an environment where participants feel comfortable expressing their views [4]. The focus group method is different from other qualitative methods in terms of objective, arrangement, and the group nature of data acquisition. A focus group study usually consists of 6 to 8 participants, who are preselected based on their similar backgrounds or experiences [4]. In addition, the discussion among the participants is focused on a specific issue to allow sufficient time to discuss it in detail. Moreover, a focus group study is led by the moderator who is responsible for raising questions for stimulating discussion by insuring a permissive and non-threatening group environment [3]. The objective is not to reach an agreement on the issue, but to expose a range of different experiences and perspectives.

By conducting a focus group study, this qualitative approach has provided an in-depth understanding of the challenges in the form of open questions and concerns related to the use of spatio-temporal trajectory datasets. The enrichment of information concerning dataset challenges with the state-of-art literature will provide a rich and detailed overview that will be of great value to researchers, practitioners, and policymakers as it provides them with information on how to effectively use trajectory datasets in various applications. The study will also contribute to the existing knowledge base on the use of trajectory datasets by providing a foundation for future research on how to effectively and ethically use this data. Ultimately, this will help to ensure that the use of

trajectory data is in line with ethical and privacy considerations and it benefits society as a whole.

The structure of the paper is as follows: The background of the study is covered in Section 2. The details of the focus group study, including the settings and participants, are outlined in Section 3. The challenges identified in using trajectory data are discussed in Section 4. The paper concludes with a summary of the findings in Section 5.

2 Background

Spatio-temporal trajectory dataset is a type of data that consists of the movement of objects. It is made up of one or more trajectories. A trajectory is a trace generated by a moving object in a geographical space represented using a series of chronologically ordered points [1, 2]. For instance, $p_1 \rightarrow p_2 \rightarrow p_3 \rightarrow \cdots \rightarrow p_n$, where each trajectory point contains a location (i.e. geographical coordinate set) and a timestamp such as $p = (x, y, t)$. A trajectory dataset typically has five trajectory data fields which are; userID (i.e. an identification of a user), date-time (a timestamp when a trajectory point is collected), longitude values, latitude values on a geographical scale, and the context (i.e. related semantics that enriches the information regarding the trajectory point). A trajectory data may also include information about the speed and direction of an object's movement [5]. The objects can be people, vehicles, animals, or any other moving entity that can be tracked. The trajectory data can be captured from several different sources [1, 2], some of them are mentioned in Table 1. Using these sources, the trajectory data is stored in a database for analysis and visualization. When plotted on a map, the spatio-temporal trajectory points can be connected to form a line or path, which shows the object's movement over time [2]. The spatial and temporal attributes can be represented using different visual encodings, such as color, size, and shape. It is also possible to represent the spatio-temporal trajectory in a multidimensional space where each dimension represents a different attribute. Spatio-temporal trajectory data is used in a variety of fields, including transportation, urban planning, environmental monitoring, marketing, and many others, to gain insights into patterns of movement and behavior [1].

3 Focus Group Study

The focus group comprised 49 individuals from diverse countries, including Denmark (DK), Netherlands (NL), Italy (IT), Germany (DE), United Kingdom (UK), Greece (GR), France (FR), Brazil (BR), and the United States (US) of America, who are experts in the field of Computer and Information Science. The participants consisted of Ph.D. students, researchers, and practitioners with a specialization in mobility data science. The focus group was moderated by Konstantinos Tserpes (Harokopio University, Greece) and featured panelists including Thierry Chevallier (AKKA Technologies, France), Bettina Berendt (Technische Universität Berlin, Germany & KU Leuven, Belgium), Franco Maria Nardini (ISTI-CNR, Italy), Luca Pappalardo (ISTI-CNR, Italy), and Mahmoud Sakr (Université Libre de Bruxelles, Belgium).

The primary objective of the focus group was to explore various challenges associated with using spatio-temporal trajectory datasets. This discussion, which lasted for two hours, was a part of the 1st International Summer School on Data Science for Mobility (http://master-school.isti.cnr.it/) held in Santorini, Greece, from 3 to 7 October 2022.

TABLE 1: SOURCES OF TRAJECTORY DATA [1, 2, 6 - 10]

No.	Types of trajectory data	Description
1	Trajectory of people	People record their movements in the form of trajectories, actively or passively, for a long time. Examples are; 1) Logging of travel routes with GPS for memorizing journeys and sharing with friends; 2) Bicycles, scooters, and runners record their paths for sports analysis; 3) On Flickr, using geotagged photos can build a trajectory as each photo has a location tag and a time stamp; 4) The “check-ins” of people on a location-based social network are treated as a trajectory when arranged sequentially; 5) Call Detail Records (CDRs) provide the mobile positioning data of users as long as they have their phones with them; 6) Credit card transactions of people hold the information of a timestamp and a merchant identification with location data; 7) Google Location History (GLH) data provides trajectory data having large temporal scales and high spatial granularity of local and international trips of people; 8) Facebook tracks the mobility data of active Facebook users using Facebook location history; 9) Surveillance cameras are used to track the movement of people and provide location data.
2	Trajectory of vehicles	1) Connected cars have multiple data sources, including lidar units, radar sensors, and cameras that record location data; 2) GPS-equipped taxis, buses, trams, vessels, and aircraft record their movements with time information. Public transportation services track this data for traffic analysis of commuters; 3) Trajectory dataset of vehicles are generated by simulating different scenarios and scenarios for testing and validation of models.
3	Trajectory of animals	Researchers in life sciences use accelerometers, gyroscopes, magnetometers, etc. sensing technologies for collecting trajectories of animals such as tigers, birds, etc. for studying animals` migratory behaviors and survival situations.
4	Trajectory of natural phenomena	Environmentists collect trajectories of natural phenomena such as tornados, hurricanes, etc. for studying and prediction of natural disasters.

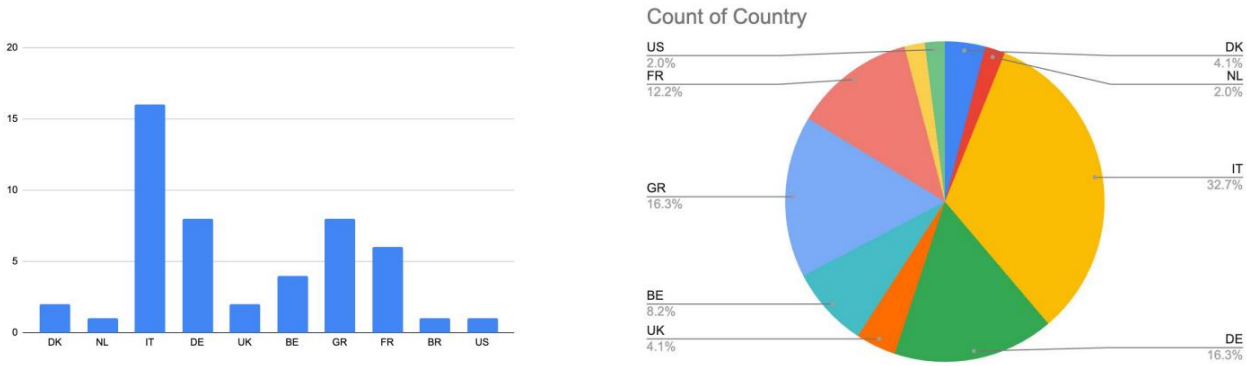

Figure 1: The diversity of participants involved in a focus group study

TABLE 2: TRAJECTORY DATA CHALLENGES

No.	Challenges in terms of issues and open research questions	Type	Concerning
1	How can we convince mobile users to share their data for a value against a contract?	Open question	Dataset construction
2	Do people come under pressure when their trajectory data is monitored?	Open question	
3	Can an idea of a digital twin be used to generate trajectory datasets?	Open question	
4	Is constructing a synthetic trajectory dataset is preferably over a real trajectory dataset?	Open question	
5	There is no benchmark exists to develop and publish trajectory datasets.	Concern	
6	How can we make trajectory datasets findable, accessible, interoperable, and reusable?	Open question	
7	There is no standard exists to preprocess the trajectory datasets.	Concern	Dataset preprocessing
8	There is no standard exists to extract spatio-temporal and kinematic features from trajectory datasets.	Concern	
9	Can we protect the data privacy by sharing the inferred knowledge than real trajectory datasets?	Open question	Dataset privacy
10	Can we use the help of social scientists to define the norms for respecting the trajectory data privacy and ethics?	Open question	
11	Can we share trajectory datasets directly to researchers rather than to companies?	Open question	Dataset sharing
12	Can downloading the trajectory dataset from the public machines helps to avoid data leaks?	Open question	
13	Can synthetic datasets overfits the trajectory prediction models for future analysis?	Open question	Dataset use
14	There is no policy exists to use a private trajectory dataset in research.	Concern	

The group's diversity and expertise in the field made for a productive and insightful conversation. The summer school was organized under the MASTER (Multiple ASpects TrajEctoRy management and analysis) project that is funded under the call H2020-MSCA-RISE-2017 to form an international and inter-sectoral network of organizations working on a joint research program to define new methods to build, manage and analyze multiple aspects of semantic trajectories. There were 14 challenges in the form of concerns and open questions raised by the participants. These challenges were categorized under the categories of trajectory dataset construction, dataset preprocessing, dataset privacy, dataset sharing, and dataset use (see Table 2).

4 Trajectory Datasets Challenges

This section presents information about the challenges in the form of open research questions and concerns associated with using spatio-temporal trajectory datasets. This information is drawn from the participants who participated in the focus group study. The information is intended to help researchers and practitioners to better understand the complexities involved in different stages of trajectory data management (i.e. dataset construction, preprocessing, privacy, sharing, and use), and to provide guidance on how to overcome these challenges. Also, the challenges related to trajectory datasets presented in this section are enriched through a review of the state-of-the-art literature [6 – 10], which ensures that the information is based on the latest research findings and advances in the field.

4.1 Trajectory dataset construction

Spatio-temporal trajectory datasets are collections of data that track the movement of objects or individuals over time and space. There are many sources for generating trajectory data, including GPS devices, RFID sensors, social media, mobile network data, vehicles equipped with sensors, and drone-based technologies [1, 2]. These sources can be used independently or in combination to generate spatio-temporal trajectory datasets. The choice of choosing data

sources depends on the nature of the research question, and the availability of data.

Challenge 1: How can we convince mobile users to share their data for a value against a contract?

The main challenge that arises regarding the construction of trajectory datasets is how to convince mobile users to share their data for research? Convincing moving users to share their trajectory data for a value against a contract can be a challenging task. However, there are several strategies found in the existing literature [11, 12] that can be used to increase the probability of success. Some of them are; 1) Transparency and informed consent: researchers need to be open and transparent about data collection and the practices that will be used for collecting data for protecting the privacy of users. For this, we need to have informed consent from users. This means providing users with clear information about the data collection, its use and giving them the choice to opt-in or opt-out. 2) Value proposition: we need to provide tangible benefits to users for sharing their trajectory data. This could include discounts, recommendations, or other services and rewards. 3) Trust and security: Build trust with users by explaining that their data will be handled securely, and the regulations for privacy and security passed by the European Union (EU)`s General Data Protection Regulation (GDPR) will comply during data acquisition [13].

Challenge 2: Do people come under pressure when their trajectory data is monitored?

Even after opting for all the strategies for convincing users to share their trajectory data and building their trust, the collection of data can put pressure on them. Users may feel that their movements are being tracked, which can make them feel uncomfortable, anxious, or even unsafe. The pressure can be amplified when the data is used for decision-making that can have a direct impact on people's lives (e.g. targeted advertising, risk assessment, or law enforcement) [14]. To mitigate this pressure, users need assurance from researchers that the data will be anonymized to protect their privacy. Furthermore, research organizations and universities should have clear data governance policies [15] in place, including data security, data sharing, and data retention, to ensure that the data is used ethically and legally.

Challenge 3: Can an idea of a digital twin be used to generate trajectory datasets?

To avoid the complex processes involved in convincing people, building their trust, and ensuring that their data is used legally, the idea of a digital twin [16] can be used for generating synthetic trajectory datasets. Digital twins, which are virtual models of physical systems, can be used for generating synthetic spatio-temporal trajectory datasets [17]. Digital twins are used for simulating the movements of users or objects in a virtual environment, and the movement data can be used for creating synthetic trajectory datasets. This scenario of data collection can be particularly useful when real-world data is not available or is difficult to collect.

Challenge 4: Is constructing a synthetic trajectory dataset is preferably over a real trajectory dataset?

The synthetic data generated by digital twins can be used for testing and validating trajectory analysis algorithms [18], as well as for training machine learning models. However, the synthetic datasets produced by digital twins may not always perfectly represent real-world conditions and may not be generalizable to real-world scenarios. Also, it's highly dependent on the quality of the model used to generate the data, and it's important to validate the synthetic data against real-world data if possible.

Challenge 5: There is no benchmark exists to develop and publish trajectory datasets.

Whether the trajectory dataset is collected from the real world or using digital twins, it should be developed and published using an existing benchmark. As per the participants of the focus group, there is no widely accepted benchmark for creating a trajectory dataset. Some possible reasons are 1) the lack of a standard format for storing trajectory data as different research groups may use different data formats, 2) the complex and diverse nature of trajectory dataset because it is collected from a wide variety of sources, and can include different types of information that makes it difficult to create a standardized dataset that can be used for a wide range of applications. Lastly, 3) lack of shared access to proprietary data as many trajectory datasets are proprietary (i.e. can only be used by the group that created them). This can make it difficult to use these datasets as benchmarks. However, there are a few datasets for trajectory data that have been developed and published in the research community can be used as benchmarks. Some examples include the Stanford Drone Dataset [19], the ETH Zurich datasets [20], and the UCY dataset [21]. Also, [22] presented an indoor crowd movement trajectory benchmark dataset. These datasets typically include data collected from drones or other mobile platforms and include information such as GPS coordinates, timestamps, and sensor measurements.

Challenge 6: How can we make trajectory datasets findable, accessible, interoperable, and reusable?

After the development of trajectory data, it is important to make it findable, accessible, interoperable, and reusable [23]. This process is important for ensuring that the data can be used by the research community for developing a wide range of applications. To make datasets findable, they should be listed in relevant data repositories, such as the Global Earth Observation System of Systems (GEOSS) [24] or the European INSPIRE Geoportal [25], and described with standard metadata fields, such as keywords, abstract, and contact information. To make trajectory datasets accessible, they should be available in an open format, such as CSV or JSON, and accessible via a web-based interface, such as a RESTful API or a web portal [26]. Additionally, the data should be accessible to a wide range of researchers, developers, and members of the public. To make

spatio-temporal trajectory datasets interoperable, they should be formatted in a way that is compatible with other datasets and systems. This can be achieved by using common data standards, such as the Open Geospatial Consortium (OGC) standards [27] for geospatial data or the ISO 19115 standard for metadata [28]. Lastly, to make trajectory datasets reusable, they should be well-documented, including information about the data collection methods, data quality, and any limitations of the data. This information should be included in the metadata in a standard format. It is important to mention that making spatio-temporal trajectory datasets findable, accessible, interoperable, and reusable is not only a technical task, but also a legal, ethical, and organizational one. Therefore, it's important to have a clear dataset-sharing agreement, data governance policies, and protocols, and to ensure compliance with laws and regulations regarding data privacy, security, and intellectual property.

4.2 Trajectory dataset preprocessing

Synthetic or real-world datasets need to pass through the process of preprocessing before using it for research [29, 30]. Preprocessing data is an important step in making it ready for analysis and can improve the accuracy, efficiency, and interpretability of the results [2, 30]. Trajectory data preprocessing can be a tedious and time-consuming task based on the application use case [30]. There are some common steps and best practices [29, 30] that are typically followed in preprocessing trajectory datasets, such as 1) data cleansing, 2) data transformation, 3) stay point detection, 4) trajectory interpolation, 5) map-matching, 6) segmentation, 7) compression, and 8) data anonymization.

Data cleansing is used for removing anomalies (i.e. caused by the poor signal of location acquisition sensors) and ensuring that the dataset is manageable, and has a unique representation of the real-world [30]. It includes format adaptation for data values, integrity constraint enforcement, deriving missing data values from the present ones, merging and removing duplicated values, and detection of outliers. More details on data cleaning can be found here [31]. Data transformation deals with converting the trajectory data into a suitable format for analysis, such as converting the raw traces into a tabular format with columns for longitude, latitude, time, or any other relevant variable [30]. A stay point detection method identifies the location (e.g. restaurant, university, etc.) where a moving person or an object has remained for a while within a specific distance threshold [29]. Trajectory data interpolation deals with the estimation of unknown trajectory values at arbitrary locations and times using the known values at locations and times [32]. The map matching process aims to map each trajectory point onto a corresponding real-world (e.g. road) segment where the trajectory point was generated [29]. Trajectory segmentation divides a trajectory into segments by spatio-temporal or semantic features for classification and clustering processes [29]. Trajectory compression is reducing the size of a trajectory, reducing the overhead related to its processing and data storage while keeping the value of the trajectory [29]. Lastly, the data anonymization process removes any information that can identify an individual to protect your privacy [30].

Challenge 7: There is no standard exists to preprocess the trajectory datasets.

The issue that was raised by the focus group concerning the preprocessing of trajectory datasets is that, in the existing literature, there exists no standard method for preprocessing trajectory datasets. The preprocessing steps and techniques employed in the literature depend on the type of data, the research question, and the intended use of the data. These steps and techniques generate results that are often varied across studies because of the variety of the datasets and the use of different definitions based on their understanding. These variations have led to a vast collection of application-specific preprocessing of trajectory datasets. Even in the case of published trajectory datasets, the details regarding preprocessing steps and techniques are missing, and the researchers need to know about the preprocessing steps which were used for generating the trajectory dataset. For instance, the study by Feng et al. [33] proposed the DeepMove model based on the deep learning model used for next-location prediction. Raw position record is taken as a location and it does not perform preprocessing. However, addressing the same problem, Urner et al. [34] and Arslan et al. [35] extracted stay points from the trajectory dataset that was acquired from the location positioning system. These examples show how the lack of a standard trajectory model definition and a common trajectory preprocessing standard limits the reproducibility and comparability of the techniques and analysis results. To address such situations, a standard need to be defined and followed for developing trajectory datasets.

Challenge 8: There is no standard exists to extract spatio-temporal and kinematic features from trajectory datasets.

During the preprocessing of trajectory datasets, the extraction of the spatio-temporal and kinematic features from trajectory datasets is also observed in existing studies [36 - 39]. Temporal features represent the features that are extracted based on temporal dimension, i.e. month, week, day, hour, etc. Semantic features describe contextual information that corresponds to the movement based on the environment. An example of a semantic feature is the point of interest visited in a building or a city. Lastly, kinematic features [29] refer to those features which deal with the motion of the object, i.e. speed, rotation, etc. It is confirmed by the focus group that there is no standard method exists for extracting spatio-temporal and kinematic features from trajectory datasets. The reason for non-existence of a standard for extracting spatio-temporal and kinematic features from trajectory datasets is because the feature extraction process entirely depends on the specific research application and the type of data. Depending on the application and the type of data, different features may be more relevant or useful than others. Therefore, it's important to choose the features that are most relevant to the task at hand.

4.3 Trajectory dataset privacy

Trajectory datasets can raise privacy concerns because they contain information about an individual's location over time [40]. This information can be used for inferring sensitive information about an individual's personal habits, routines, and relationships [41]. For instance, knowing about an individual's home or work location, or to track an individual's movements in real-time, or inferring individual's personal habits such as their religious or political beliefs, their health status, or their social connection [41, 42]. In addition, anonymized trajectory data may still contain enough information to allow for re-identification of individuals, especially if the data is combined with other sources of information [43]. Moreover, trajectory data can be used to conduct surveillance on individuals or groups, which can have negative consequences for civil liberties and human rights.

Challenge 9: Can we protect the data privacy by sharing the inferred knowledge than real trajectory datasets?

To avoid privacy-related issues, the focus group has suggested that sharing inferred knowledge from spatio-temporal trajectory datasets rather than the raw data itself can help to protect data privacy. By inferring knowledge from the data, sensitive information such as individual identities and specific locations can be removed. However, it's important to note that the level of privacy protection provided by inferred knowledge will depend on the methods used and the level of detail included in the inferred knowledge. It's possible that certain types of inferred knowledge, such as aggregate statistics or patterns, could still reveal sensitive information about individuals or groups [44]. Moreover, sharing inferred knowledge may not be sufficient for protecting data privacy in all cases. For example, if the inferred knowledge is combined with other information sources, it could potentially be used to re-identify individuals or groups in the dataset [44]. Furthermore, if the methods used for inferring the knowledge are not robust to attacks, it could be possible for an attacker to reverse engineer the original data. Ultimately, the best way to protect data privacy when sharing trajectory datasets will depend on the specific dataset and use case, and should be evaluated on a case-by-case basis. It is crucial to consult data privacy and security experts for determining the best approach.

Challenge 10: Can we use the help of social scientists to define the norms for respecting the trajectory data privacy and ethics?

Apart from data privacy and security experts, social scientists can also play a crucial role in addressing ethical and privacy concerns related to trajectory datasets [44]. They can help in identifying potential ethical issues and suggest ways to address them, as well as understand the social and cultural factors that influence people's perceptions of data collection and use. Additionally, social scientists can help organizations establish fair and responsible norms and protocols for data governance, considering the perspectives of all stakeholders.

4.4 Trajectory dataset sharing

Trajectory dataset sharing refers to the practice of making trajectory datasets available to the public or scientific community [2]. By sharing these datasets, researchers and developers can use these datasets and gain insights for developing new research applications. Also, sharing datasets promote openness and transparency in research, enabling others to verify results.

Challenge 11: Can we share trajectory datasets directly to researchers rather than to companies?

The question that was raised by the focus group is that whether sharing trajectory datasets directly to researchers than to companies is good practice? In fact, this practice can be used to avoid commercial gain. Researchers are often interested in studying or testing specific hypotheses and may not have the same commercial interests as companies. Sharing the datasets directly with researchers can help ensure that the data is used for scientific and academic purposes, rather than for commercial gain. It's also important to note that sharing spatio-temporal trajectory data directly with researchers can be beneficial for the company or organization that owns the data. Researchers often generate insights and new knowledge that can be used to improve products, services, and policies, leading to benefits for the company or organization. When sharing data with researchers, it is important to consider the data privacy and security risks. The dataset should be de-identified to protect the privacy of individuals. Additionally, the data should be shared through a secure platform, such as a secure data repository or a virtual private network, to ensure that the data is not accessed by unauthorized parties. However, it's important to have a clear data sharing agreement that outlines the terms and conditions of data sharing, including the purpose of the research, how the data will be used, how the data will be protected, and how the results of the research will be shared.

Challenge 12: Can downloading the trajectory dataset from the public machines helps to avoid data leaks?

The focus group suggested that downloading spatio-temporal trajectory datasets from public machines, such as publicly available data repositories or open data portals, can help in reducing the risk of data leaks. When datasets are publicly available, the risk of unauthorized access is lower, as the data is accessible to anyone who knows where to find it. Public data repositories and open data portals typically have security measures in place to protect the data from unauthorized access, such as firewalls and intrusion detection systems [45]. However, it's important to note that downloading datasets from public machines of universities or research institutes does not completely eliminate the risk of data leaks. The datasets may still contain sensitive information, such as personal identifiers, that could be used to re-identify individuals. It's also important to check the terms of use, privacy policy and the data sharing agreement of the data repository to make sure that the data is being shared legally and ethically. Moreover, the use of public machines for data storage and sharing is just one aspect of data security. Other measures such as access controls, encryption, and monitoring plans

are also important for ensuring data security and preventing data leaks.

4.5 Trajectory dataset use

There are several challenges exists when exploiting trajectory datasets for different research applications. Some of them are; data sparsity (i.e. there are many gaps in the data, making it difficult to accurately infer movement patterns), data quality (i.e. data can be noisy and may contain errors), scalability (processing large-scale trajectory data can be computationally expensive and time-consuming), data integration (i.e. the data from different sources may be in different formats, have different temporal and spatial resolutions, and use different coordinate systems), and data visualizations (i.e. trajectory data often involves large amounts of data and multiple dimensions). Their details can be found in the literature [1, 2, 30] as they are very well-documented and investigated by researchers.

Challenge 13: Can synthetic datasets overfits the trajectory prediction models for future analysis?

The focus group is concerned about the use of synthetic trajectory datasets. According to them, overfitting is a critical concern has been observed in their past experimentations when using synthetic trajectory datasets for training and testing trajectory prediction models. Overfitting occurs when a model is trained too well on the training data and performs poorly on new, unseen data [46]. This can happen if the synthetic dataset is not representative of the real-world conditions the model will be used in, or if the model is too complex for the amount of data available.

Existing literature [46] suggests different ways for ensuring that synthetic trajectory datasets do not overfit the prediction models. For instance, when creating synthetic datasets, it is important to confirm that the dataset is representative of the real-world conditions the model will be used in. This will ensure that the model is not overfitting the training data. In addition, the application of cross-validation (i.e. where the dataset is split into multiple subsets and the model is trained and evaluated on each subset) can also be explored for minimizing the chances of overfitting the model. Moreover, the techniques of regularization can help to prevent overfitting the models by adding a penalty term to the model's cost function. More details on regularization techniques can be found here [47]. Ultimately, the use of simpler models with fewer parameters can also help to reduce the risk of overfitting.

Challenge 14: There is no policy exists to use a private trajectory dataset in research.

To avoid the problem of overfitting the models with synthetic datasets, the exploitation of private real-world trajectory datasets is also observed in the existing literature [48]. However, as per the focus group, there is no widely accepted policy exists for using private trajectory datasets in research applications. There are some reasons why such policy doesn't exist. For example, trajectory datasets usually contain sensitive information. It is important for researchers to be aware of the data usage agreements and legal requirements. In some cases, researchers may require ethical clearance with an institutional review board before using private trajectory datasets. This makes it complex for creating a standard policy that would balance the need for data privacy with the need for data access for research. In addition, creating a policy for using private trajectory datasets in research is a complex task that requires significant resources (e.g. legal expertise, data privacy expertise, and knowledge of the research community). Lastly, there is no standard for collecting, storing, and preprocessing trajectory data. This makes it complex for creating a policy that would be applicable to a wide range of private datasets.

5 Conclusion

The rapid advancement in collecting spatio-temporal trajectory data has brought numerous benefits to various industries, including urban planning, telecommunication optimization, and human behavior analysis. Despite its potential, the use of trajectory datasets also faces challenges which are described in this article. To better understand these challenges, this study employed a focus group discussion with a diverse group of participants from academia and industry. The focus group study, along with a literature review, provided an in-depth understanding of the issues and concerns related to the use of spatio-temporal trajectory datasets. The findings will be of great value to researchers, practitioners, and policymakers, contributing to the existing knowledge base and ensuring the ethical and responsible use of this data for the benefit of society.

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